

EDUCATION

Miami University

- Bachelors of Science
- Mechanical Engineering

Oxford, OH
Expected Graduation 2029
3.51 GPA

Saint Xavier High School

- Honoree of National Honors Society
- Honoree of Tiger Mentor Program
- Honoree of KHSAA Academic All State
- Honoree of KHSAA All State Team Awarded Tennis Coaches Award
- Awarded Alan Donhoff Award for Character & Leadership
- Awarded the Tennis Coaches Award
- Trace of Hope Scholarship Recipient
- 2019 USTA Southern Family of the Year
- Principles List Junior & Senior Year
- Honor Roll Freshman & Sophomore Year
- Trace of Hope Scholarship Recipient
- Completed AP course in English Language & Composition
- Completed AP course in US Government & Politics

Louisville, KY
Graduated 2025
3.9 GPA

RELEVANT EXPERIENCE

Louisville Country Club

Assistant Tennis Coach

- Led individual and group training sessions by adjusting instruction to each player's strengths to improve overall performance.
- Built strong relationships with families and players to keep them engaged, motivated, and progressing throughout the season.
- Guided junior coaches by sharing feedback and communication strategies during practices and matches.
- Worked closely with other coaches to improve training plans and day-to-day operations.

Louisville, KY
June 2025 - August 2025

Kentuckiana Pool Management

Lifeguard

- Maintained a safe pool environment by consistently monitoring swimmers and enforcing safety rules.
- Handled emergency situations using CPR and first aid training while staying calm under pressure.
- Helped train new lifeguards on safety procedures and expectations for interacting with patrons.
- Checked pool facilities and equipment to ensure safe operating conditions.

Louisville, KY
June 2023 - August 2025

Louisville Youth Advocacy Board

Louisville, KY

President

- Led board initiatives focused on increasing youth involvement and strengthening community outreach.
- Worked with local organizations to expand opportunities and resources for youth programs.
- Supported and mentored board members to improve leadership and teamwork.
- Organized student led fundraising events that raised money to provide meals for children in need and increased community participation

August 2024 - May 2025

COURSES & PROJECT EXPERIENCE

CEC 112: Imagination Ingenuity Impact 2

Wind Turbine Project

February – April 2026

- Designed, built, and tested a functioning wind turbine, evaluating how engineering design decisions affected electrical power generation.
- Used MATLAB and Arduino to collect, analyze, and visualize turbine performance data while troubleshooting electrical wiring and sensor issues.
- Conducted controlled experiments to optimize blade angle, blade material, blade quantity, and gear ratios, identifying a 20° blade angle with four flat blades as the highest-performing configuration.
- Evaluated turbine efficiency by comparing theoretical and experimental power output and using data-driven testing to improve overall system performance.
- Collaborated on the development of a team website using HTML and CSS to document the engineering design process and project results.

Personal Project

Acebot QD001 + QD007

June – July 2026

- Built and integrated an ESP32-based robotic car with a 4-DOF robotic arm.
- Calibrated servo motors and configured embedded firmware for full system operation.
- Diagnosed and resolved hardware, wiring, and software integration issues.
- Tested and optimized Bluetooth-controlled vehicle and robotic arm functionality.

CEC 111: Imagination Ingenuity Impact 1

Fluid Systems Design Project

October 2025

- Led the redesign of a fluid system by simplifying components, improving reliability and ease of assembly
- Adjusted designs based on available materials, reducing valve count by roughly one-third while maintaining performance.
- Worked in defined team roles to keep testing and redesign organized and efficient.
- Tested multiple system iterations to identify leaks, pressure buildup, and flow issues, leading to steady performance improvements.

CEC 111: Imagination Ingenuity Impact 1

Electronic Monitoring System Project

October 2025

- Built an electronic monitoring system using Arduino, resistors, LEDs, and breadboards to detect fluid flow.
- Wrote and refined Arduino code to respond to changes in the system during testing.
- Troubleshoot circuits using basic circuit analysis to improve reliability.
- Collaborated with teammates to test, debug, and improve the final system.

CEC 111: Imagination Ingenuity Impact 1

Spectrometer (Spectrophotometer) Project

November 2025

- Built a working spectrophotometer to measure absorbance across different light spectra.
- Calibrated the device using MATLAB and generated spectral images for analysis.
- Created a standard calibration curve to compare absorbance values across samples.
- Reviewed results to improve accuracy and identify sources of error.

CEC 111: Imagination Ingenuity Impact 1

App Prototyping & Data Analysis Project

November 2025

- Led the design of an ancestry-focused app that allowed users to find and compare population data by ancestry across U.S. states.
- Designed app prototypes in Figma with an emphasis on intuitive navigation and clear data presentation.
- Analyzed ancestry population datasets using Excel and Tableau to identify trends and regional differences.
- Created visualizations that made complex demographic data easy to understand and compare.

ACTIVITIES

Kappa Alpha Order – Epsilon Lambda Chapter

Member

Oxford, OH

January 2026 - Present

- Successfully completed the fraternity's new member education and was initiated into the fraternity.
- Participated in chapter meetings, philanthropy events, and brotherhood activities.
- Worked with fellow members to support service projects, campus involvement, and professional development.

Miami University Club Tennis Team

Member

Oxford, OH

August 2025 - Present

- Competed in practices and matches while balancing academics at the collegiate club level.
- Collaborated with teammates to maintain a competitive and supportive team culture.

Blessings in a Backpack

Mission Ambassador/Volunteer

Louisville, KY

August 2019 - May 2025

- Assisted at frequent charity events, helping connect donors, volunteers, and community members to the cause
- Spoke with community members at events to raise awareness about food insecurity
- Helped lead outreach efforts that increased support for families in need
- Coordinated volunteers to ensure food was distributed efficiently

Project Warm

Volunteer

Louisville, KY

December 2023 - January 2025

- Worked on a team to winterize homes for families in need
- Assisted with installing insulation and sealing windows and doors.
- Helped improve home safety and comfort during winter months

Saint Xavier High School

Varsity Tennis Player

Louisville, KY

August 2021 - May 2025

- Competed at the varsity level in a highly competitive program, demonstrating discipline and resilience.
- Supported team leadership by setting a positive example during practices and matches.
- Named captain for Junior and Senior year

Xavierian Leadership Institute

Participant

Louisville, KY

August 2024 - May 2025

- Engaged in leadership development focused on service, self-awareness, and ethical decision-making.
- Participated in discussions and activities that strengthened communication and leadership skills.

Tiger Mentor Program

Mentor

Louisville, KY

August 2023 - May 2025

- Mentored younger students by offering guidance on academics, athletics, and school involvement.
- Helped foster a welcoming and supportive school community.

Saint Xavier High School

Football Player

Louisville, KY

June 2021 - December 2025

- Developed teamwork, accountability, and discipline through a structured team environment.
- Adapted quickly to new systems and expectations in a physically demanding sport.

SKILLS / CERTIFICATIONS

<u>Technical Skills</u>	<u>Software</u>	<u>Design Tools</u>	<u>Communication</u>	<u>Certifications</u>
• Basic Circuit Analysis • Fluid Systems Testings • Data Analysis • System Troubleshooting	• MATLAB • Claude Code • Arduino • Excel • Tableau • Figma	• Breadboards • Electronic Sensors • LEDs • Prototype Testing	• Technical Communication • Team Collaboration • Public Speaking • Mentoring • Project Coordination	• CPR • First Aid • AED • Lifeguard Certification